Fiscal Year:	FY 2007	**Task Last Updated:**	FY 12/19/2007
PI Name:	Mitchell, Cary Ph.D.		
Project Title:	(NSCORT) for Advanced Life Support at Purdue/Howard/AAMU - Food Safety and Food Processing		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	Purdue University		
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City:	West Lafayette	**State:**	IN
Zip Code:	47907-2010	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSCORT
Start Date:	12/01/2002	**End Date:**	11/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	5
No. of Master's Candidates:	9	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date is 11/30/2006, per info from PI office (2/2009)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Applegate, Bruce (Purdue University) Mauer, Lisa (Purdue University) Williams, Leonard (Alabama A&M University)		
Grant/Contract No.:			
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Performance Goal Text:			

Task Description:	Purdue and Alabama A&M Universities continue to address key components associated with: food processing, food safety, energy requirements, and long term exposure of food to radiation. Research to understand the effects of radiation on various food stuffs has been continued, characterizing the effects of space-relevant doses of gamma-radiation on select foods, food ingredients, and dietary supplements. Threshold radiation levels for quality changes were also identified. Radiation can initiate a process known as autoxidation, producing a lipid free radical, which reacts with these products leading to off odors and flavors associated with rancid fats. Studies using soybean and peanut oils demonstrated that the addition of antioxidants to irradiated oils reduces total oxidation and delays maximum oxidation from 7 to 42 months for soybean oil and from 21 to 49 months for peanut oil. The impact of radiation on proximate and nutrient profiles and antioxidant capacities of carrots, tomatoes, strawberries, and apples was examined. Moisture, ash, protein, total sugar, water activity, calcium, magnesium, sodium, and potassium were relatively unaffected by radiation, however, total extractable fat, vitamin C, carotenoids, and lycopene were affected by radiation. To ensure food safety and shelf life of potential salad crops, research is being performed to examine the effect of various physical and chemical control strategies. Synergistic effects using the combination of chemical and physical methods showed the greatest benefit. Research is also continuing to identify potential routes of contamination and effective control strategies using an integrated HACCP (hazard analysis and critical control points) expanded to evaluate the ALS system as a whole for identification of key parameters and possible points of contamination from the various waste processing regimes, hydroponics, and tilapia waste water. Pathogen detection using recombinant phage has progressed. Focus of the current research is the purification of the phage utilized prior to detection. This is crucial to reduce false positives. Collaboration with Embedded Technologies has provided a potentially significant alternative to the integrated pathogen detection food packaging format. It was successfully demonstrated that antibodies could be immobilized by embedding them in plastic with their technology. This could be used in numerous formats to hasten pathogen detection.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Further development of nano-scale detection of pathogens in packaged foods Development of novel food products and technologies for vegetarian and health-food markets Examine the effect of various physical and chemical control strategies to increase shelf life of foods. Increase understanding of radiation on wheat , oils and vegatables.
	Pathogenic detection using phage display Salmonella spp bacteriophage: To facilitate the construction of a modified P22 bacteriophage for Salmonella spp. it was necessary to construct a recombination vector for insertion of a modified tail spike protein. The recombination vector was constructed by removing approximately 1 kb of the P22 genome in the region of orf 201 and orf 80 to allow the insertion of DNA for recombination. A multicloning site and a TA cloning site were inserted to facilitate rapid insertion of modified DNA to construct the appropriate epitope. Primers were utilized to amplify the tailspike protein from P22 with the appropriate His modifications. The preparative host strain repressor gene cassette was constructed for insertion into the preparative host strain genome. Work is continuing on inserting the modified tailspike gene in the previously constructed P22 recombination vector. Appropriate promoter and terminator configurations were added to the modified tailspike protein to allow repression of expression of the His modified tailspike protein in the preparative host strain. Preparative host strain containing the lacI repressor gene cassette was constructed. E. coli O157:H7 bacteriophage: The apparent host specificity of phiV10, its ability to lysogenically convert without transducing host genes, and its lack of identifiable virulence determinants suggest that phiV10 should be useful in biotechnology and that its use in biotechnology should not constitute any undue hazard, making it a good candidate for detection assays. Listeria monocytogenes bacteriophage: An initial screening of a Listeria monocytogenes library to isolate a lysogenic bacteriophage producing preliminary results that suggest success and we are currently further evaluating the isolated bacteriophage and continuing to screen the library. Packaging: The initial proposed product would contain a lateral flow assay for the multiplexed determination of multiple pathogens. We have begun collaboration with Embedded Concepts, which has patented technology which allows the embedding of antibodies into plastics. This innovation would greatly enhance the potential for integration of the phage based assay into food packaging material without the need foe lateral flow simplifying development of stand alone kits involving sample preparation and detection all in one. FOOD PROCESSING AND PACKAGING Research efforts for 2005-2006 focused on: 1) characterizing the effects of space-relevant radiation doses of gamma-radiation on select foods, food ingredients, and dietary supplements and identifying threshold radiation levels for quality changes, and 2) characterizing the impact plant cultivars and growth conditions have on nutrient profiles, antioxidant capacity, functionality, and acceptability of foods. Radiation Effects on Oils Radiation can initiate a process known as autoxidation, where an initiator species reacts with the lipid, removes a hydrogen, and produces a lipid free radical. This lipid free radical can then react with other fatty acids and produce an accelerating chain reaction of similar events during propagation. Products of this oxidation of fatty acids include peroxides, alcohols, aldehydes, and carbonyls. Further reactions with these products lead to off odors and flavors accustomed to rancid fats. Studies using soybean and peanut oils (with and without added antioxidant, irradiated at 0, 3, 10, 100, 200, 400, 600, 800, and 1000Gy, and stored at 65ºC) found that addition of 0.02% TBHQ (antioxidant) reduces total oxidation and delays oxidation maxima from 7 months to 42 months for soybean oil and from 21 to 49 months for peanut oil. Radiation Effects on Antioxidants Radiation doses during transit and on the surface of Mars are expected to be as high as 3 sieverts (approx. equivalent to

Task Progress:

3 Gy), excluding solar events. Studies have shown that antioxidants may provide long term health protection from oxidative stress caused by radiation exposure; therefore, to counteract the impact of elevated radiation exposure on astronaut health, consumption of antioxidants will be important. In addition to vitamin supplements, antioxidants within foods will also provide protection. While radiation dose significantly affected the stability of vitamin C in solution, stability of vitamin C in dry formulations (vitamin C supplements, multivitamins, and ascorbic acid powder) was stable following radiation doses of up to 1000 Gy.

Radiation Effects on Fruits and Vegetables

Determination of appropriate cultivars for not only growth but also food use are important to the success of a sustainable human habitat. Preliminary studies have shown that cultivar selection and growth conditions impact the composition and acceptability of crops such as sweet potato and carrot. While moisture, ash, protein, total sugar, water activity, calcium, magnesium, sodium, and potassium were relatively constant upon exposure to all radiation doses, total extractable fat, vitamin C, carotenoids, and lycopene were significantly affected by radiation doses. The finding that radiation exposure may increase the extractability of fats from plant tissues could be utilized for further food processing development and commercial applications.

Radiation Effects on Wheat

Wheat is the most studied candidate crop and has the highest priority. Apogee and Perigee cultivars of wheat were developed at Utah State University to yield high amounts of wheat berries with a minimum amount of inedible crop waste. While growth conditions have been characterized, further analysis of these cultivars must be conducted to characterize the protein, lipid, and starch functionalities related to food quality, as well as antioxidant capacities, and to compare the food-functionality of these cultivars to common cultivars used in the food industry. Because wheat can be shipped as a bulk ingredient and/or grown in space, additional studies investigated the effects of radiation exposure on the quality and functionality of wheat. For a Mars mission, results indicate that Apogee is a more suitable candidate cultivar than Perigee due to its overall stability and nutritional properties.

OPTIMAL FOOD SAFETY IN ADVANCED LIFE SUPPORT

A. Inactivation of Salmonella by pulsed light sterilization and sanitizers. The 1% peracetic acid solution was the most effective treatment while the 2% commercial sanitizer was the least effective sanitizer over 8 days of storage. Overall, 1% peracetic acid was the most effective in reducing the populations Salmonella spp. on the surface of whole tomatoes.

Pulsed Light and Sanitizer Treatment. Several preliminary studies conducted in our laboratory with tomatoes aided in the determination of a suitable distance from the light source and the number of pulses that could be used without a significant temperature increase. After 2 and 8 days of storage, 2.02 to 4.54, 0.80 to 2.10 and 5.76 to 4.86 log CFU/ml of Salmonella was recovered from the surface of the tomatoes treated with a combination of pulsed-uv light and 1% hydrogen peroxide, 1% peracetic acid and 2% commercial sanitizer, demonstrating the use of pulsed-uv light in combination with sanitizers is highly effective in inactivating Salmonella on surface of whole tomatoes. Results also indicated the use of pulsed-uv light alone wasn't as effective in significantly inactivating Salmonella on the surface of whole tomatoes compared to the combination and control treatments. B. Microbial ecology of water from hydroponic distribution system. The development of bacterial communities in hydroponic water distribution systems leads to a food chain which supports the growth of microorganisms incompatible with water quality requirements and esthetics. In this study, our hypothesis was that most plant or waterborne microorganisms can thrive in a hydroponic system during growth of salad crops, due to the abundant of nutrients present in the water and generally created by the plants. There was a significant decline ($p = 0.05$) in the numbers of mesophilic microorganisms in the hydroponic system over 12 weeks of the experiment.. The reduction was sustained after 7 weeks. After seven weeks, Pseudomonads were the only bacteria detected by traditional plating methods.

C. Pre-harvest intervention for L. monocytogenes on carrot seeds. No Listeria monocytogenes was detected in uninoculated carrot seeds. Some cells that adhered to the seeds during dipping process may have been killed during drying and storage before chlorine, hydrogen peroxide and heat treatment. Researchers have suggested the process of drying seeds after inoculation could, in effect, protect bacteria that entered the seeds through cracks and crevices against inactivation by chlorine. It's suggested that when active chlorine comes into contact with an environment with high levels of organic matter, such as carrot seeds, the potency is quickly diminished.

This very likely could have happen in this experiment, resulting in cells surviving chlorine concentrations as high as 0.1% in dip solutions to inactivate L. monocytogenes that infiltrated the seeds. With a scanning electron microscope, L. monocytogenes biofilms, an adhesive and threadlike substance surrounding the bacterial cells, was observed on the inoculated carrot seeds after 7 days of storage at 8 °C. There was no reduction in the Listeria populations on the carrot seeds soaked at 21°C (control) for 5 or 10 min. During the 7 and 21-day storage, L. monocytogenes within the liquid contiguous to the surface of the carrot seeds developed biofilms, making L. monocytogenes more resistant to the disinfectants and heat treatments. The increase in CFU per milliliter of dip solution and CFU per gram of seed appearing in 10 min and 2 min treatment for chlorine and hydrogen peroxide, when compared with the control dip (water) suggested that longer exposure in peptone solution also facilitates release of L. monocytogenes from the seeds.

SUPPLEMENTAL INFORMATION (provided in 2007):

BRUCE APPLEGATE, PURDUE UNIVERSITY: Pathogenic detection using phage display Salmonella spp. bacteriophage: To facilitate the construction of a modified P22 bacteriophage for Salmonella spp., it was necessary to construct a recombination vector for insertion of a modified tail spike protein. The recombination vector was constructed by removing approximately 1 kb of the P22 genome in the region of orf 201 and orf 80 to allow the insertion of DNA for recombination. A multicloning site and a TA cloning site were inserted to facilitate rapid insertion of modified DNA to construct the appropriate epitope. Primers were utilized to amplify the tailspike protein from P22 with the appropriate His modifications. The preparative host strain repressor gene cassette was constructed for insertion into the preparative host-strain genome. Work continues on inserting the modified tailspike gene in the previously constructed P22 recombination vector. Appropriate promoter and terminator configurations were added to the modified tailspike protein to repress expression of the His-modified tailspike protein in the preparative host strain. Preparative host strain containing the lacI repressor gene cassette was constructed.

E. coli O157:H7 bacteriophage: The apparent host specificity of phi V10, its ability to lysogenically convert without transducing host genes, and its lack of identifiable virulence determinants suggest that phi V10 should be useful in biotechnology and that its use in biotechnology should not constitute any undue hazard, making it a good candidate for

detection assays.

Listeria monocytogenes bacteriophage: An initial screening of a Listeria monocytogenes library to isolate a lysogenic bacteriophage producing preliminary results that suggest success and we are currently further evaluating the isolated bacteriophage and continuing to screen the library.

Packaging: The initial product includes a lateral-flow assay for the multiplexed determination of multiple pathogens. We have begun collaboration with Embedded Concepts, which has patented technology allowing the embedding of antibodies within plastics. This innovation greatly enhances the potential for integration of the phage-based assay into food packaging material without the need for lateral flow, thereby simplifying development of stand-alone kits involving sample preparation and detection integrated into one product.

LISA MAUER, PURDUE UNIVERSITY: FOOD PROCESSING AND PACKAGING

Initial research efforts focused on 1) characterizing effects of space-relevant radiation doses of gamma-radiation on select foods, food ingredients, and dietary supplements and identifying threshold radiation levels for quality changes; and 2) characterizing the impact of plant cultivars and growth conditions on nutrient profiles, antioxidant capacity, functionality, and acceptability of foods.

Novel Packaging, Processing and Effects of Radiation on Wheat, Fruits and Vegetables.

The intake of fresh food is very important to a crew's welfare, and thus its quality can have a tremendous impact on a crew's morale and the success of a long mission. Radiation affects the nutritional content, stability, and shelf life of foods. The objectives of this study were to 1) investigate the effects of low-dose gamma- radiation and time on specific nutrients and quality traits of fresh apples, strawberries, carrots, and tomatoes 2) investigate the effects of low-dose gamma-radiation and time on specific nutrients and quality traits of different cultivars of fresh carrots and tomatoes, and 3) investigate the effects of low-dose gamma-radiation, time, and temperature on specific nutrients and quality traits of freeze- dried and air-dried apples, strawberries, carrots, and tomatoes. The fresh and processed fruits and vegetables were exposed to levels of gamma- radiation ranging from 0Gy to 10000Gy (1Gy = 1Sv) using a Gammacell 220. The fresh fruits and vegetables were stored at 25ºC for 1 and 3 days while the processed fruits and vegetables were stored at 25ºC and 35ºC for 6 months. After the different storage times, changes in the levels of specific macro- and micro-nutrients, quality traits, and sensory evaluations were done. For the fresh fruits and vegetables after day 1, macro-nutrients such as protein, carbohydrate, sugar, calcium, magnesium, sodium, and potassium were relatively constant upon exposure to radiation doses of up to10kGy. At low radiation doses (1-5Gy) there was no significant effect on the antioxidant capacity, total phenolics, ascorbic acid (vitamin C), carotenoids (a, ß, lycopene). However, at higher radiation doses (= 10Gy), strawberries and tomatoes were more vulnerable than carrots and apples (i.e. breakdown of antioxidant capacity, total phenolics, ascorbic acid (vitamin C), and lycopene was observed). Rotting of the fruits and vegetables was observed after day 3.

Radiation Effects on Oils

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For the oils and peanut-stability project, the impact of space-relevant radiation doses on peanut and soybean oils (with and without antioxidants) as well as effects of radiation on intact peanuts is being finalized.

A manuscript is being developed on effects of space-relevant doses of gamma radiation on wheat. Gamma-radiation

	effects have been characterized on wheat macronutrients, antioxidant capacity, functionality in breads, and sensory profiles of breads.
Bibliography Type:	Description: (Last Updated: 04/16/2020)
Articles in Peer-reviewed Journals	Perry L, Shroyer M, Minocha U, Farris L, Applegate B. "Genomic analysis of Escherichia coli O157:H7 bacteriophage phiV10." Journal of Bacteriology, In review, 2006. , Jul-2006
Articles in Peer-reviewed Journals	English T, Austin N, Williams LL. "Inactivation of Salmonella spp. by pulsed light sterilization. " Submitted. Journal of Food Science and Technology, July 2005. , Jul-2005
Articles in Peer-reviewed Journals	English T, Austin N, Williams LL. "Efficacy of commercial sanitizers of reduction of Salmonella on whole tomatoes." In press. J. Food Prot., 2005. , Jan-2005
Papers from Meeting Proceedings	Williams LL, Dunham J, Walker L. "Validations of selected disinfectants and heat treatments as critical control points inactivating L. monocytogenes on carrot seeds. " International Conference on Environmental Systems, July 2006. International Conference on Environmental Systems, Paper #2069, July 2006. , Jul-2006
Patents	Filed February 2004. Provisional patent, 2004. Feb-2004 Applegate B, Ladisch M. "Phage Detection of Pathogens (PDP System)."